P-A-1056B
February 24, 1976
SUPERSEDING
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April 10, 1969

FEDERAL SPECIFICATION

ABSORBENT MATERIAL, OIL AND WATER (FOR FLOORS AND DECKS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type and grade of absorbent material for use in absorbing oil and water from floors and decks (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-S-48 - Sacks, Shipping, Paper.
VV-L-820 - Lubricating Oil, General Purpose (Light).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20406.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specification, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extend specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 7930

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American Society for Testing and Materials (ASTM) Standards:

C 431-65 Standard Methods for Sampling and Evaluation of Sorptive Mineral Products Used as Floor Absorbents.

(Application for copies should be addressed to the American Society for Testing and Material, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classificationn Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. The absorbent material shall consist of a uniform mixture of minerals of the silicate type processed in such a manner as to conform to the characteristics and requirements of table I when tested as specified in 4.3.4.

TABLE I - Characteristics

Characteristic	Maximum	Minimum
Matter retained on a No. 6 U.S. Standard sieve	1.0 percent	0.0
Matter retained on a No. 30 U.S. Standard sieve	99.0 percent	52.0 percent
Matter retained on a No. 40 U.S. Standard sieve	99.8 percent	73.0 percent
Matter retained on a No. 60 U.S. Standard sieve	100.0 percent	90.0 percent
Ability to absorb lubricating oil (per gram of sample)	----------	0.6 ml.
Ability to absorb distilled water (per gram of sample)	----------	0.7 ml.
Solubility in distilled water	1.5 percent	----------

3.2 Resistance to attrition. When tested as specified in 4.3.4, not more than 10 percent of the absorbent material shall pass through an 80 mesh screen.

3.3 Workmanship. The absorbent material shall be clean, uniform, and free from lumps or foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter. For purposes of sampling, an inspection lot for examinations and tests shall consist of material all from the same manufactured batch.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.3.3. Each lot of end item shall be examined for visual, net weight, and preparation for delivery defects. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled unit containers of the same capacity, consisting of absorbent material manufactured in one batch, for the examinations in 4.3.1.1 and 4.3.1.2, in units of shipping containers for examination in 4.3.2.1, and in units of pallets for the examination in 4.3.2.2.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one filled unit container of absorbent material.

Examine	Defect
Contents (workmanship)	Not clean, not homogeneous. Lumpy. Presence of foreign matter.

4.3.1.2 Examination for net weight. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net weight per container, for all sample units examined, is less than required.

4.3.2 Examination of preparation for delivery.

4.3.2.1 Examination for packing and marking. An examination shall be made to determine compliance with the packing and marking requirements. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. Inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery.

Examine	Defect
Container	Not as specified.
Marking	
Material	Component missing or damaged
Workmanship	Distortion of containers

4.3.2.2 Examination of closure. When shipping containers are required to comply with UU-S-48, the examination for defects in closure shall be in accordance with that specification.

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4.3.3 Inspection levels and acceptable quality levels (AQL's) for examination. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL's
4.3.1.1	I	2.5
4.3.1.2	S-2	N.A.

4.3.4 Testing of the end item. Each lot of end item shall be tested for the characteristics specified in table I in accordance with test methods specified in table IV. The lot size shall be expressed in pounds of absorbent material manufactured in one batch. The sample unit for testing shall be a 2-pounds composite obtained by combining equal portions from samples selected at random throughout the lot in the frequency indicated in table III. The composite shall be placed in a clean, dry, glass, sealed container. Care shall be exercised to prevent contamination or alteration of the absorbent material during the sampling, compositing, storage and testing. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if the composite fails to meet any test requirements specified.

TABLE III. Sampling for end item testing

Lot size (pounds)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. Instructions for testing of the end item

Characteristic	Specification reference		Requirements applicable to	Number determinations per unit	Results reported as	
	Requirement	Test method	Composite sample		Pass or fail	Numerically to nearest
Sieve test:						
No. 6 U.S. Standard sieve	Table I	4.4.1	X	1	X	
No. 30 U.S. Standard sieve	Table I	4.4.1	X	2		Average of 2 determinations to nearest 0.5 percent
No. 40 U.S. Standard sieve	Table I	4.4.1	X	2		Average of 2 determinations to nearest 0.5 percent
No. 60 U.S. Standard sieve	Table I	4.4.1	X	2		Average of 2 determinations to nearest 0.5 percent

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TABLE IV. Instructions for testing of the end item (con.)

Characteristic	Specification reference		Requirements applicable	Number determinations per/unit	Results reported as	
	Requirement	Test method	Composite sample		Pass or fail	Numerically to nearest
Absorption of lubricating oil	Table I	4.4.2	X	2		Average of 2 determinations to nearest 0.1 ml.
Absorption of distilled water	Table I	4.4.3	X	2		Average of 2 determinations to nearest 0.1 ml.
Solubility in distilled water	Table I	4.4.4	X	2		Average of 2 determinations to nearest 0.1 percent
Resistance to attrition	3.2	4.4.5	X	1		

4.4 Test methods.

4.4.1 Sieve test. Weigh 200 ± 1.0 g. of the absorbent material and transfer to the coarsest sieve. Determine matter retained on each sieve by testing as specified in ASTM C 431-65.

4.4.2 Absorption of lubricating oil. Tape an 18-mesh screen over one end of a glass tube measuring 9 inches in length and 30 mm. in inside diameter. Pour 50 g. of the sample into the open end of the glass tube held in vertical position. Fine particles passing through the screen shall be collected and returned to the top of the tube. The glass tube shall then be set at an angle of 60° to the horizontal, above a 100-ml. graduate, and 75 ml. of a lubricating oil conforming to grade 10 of VV-L-820 at 25° ± 2°C. shall be poured into the open end. The oil is absorbed by the sample until the saturation point is reached, at which time the surplus oil drains into the receiving graduate. Allow a draining period of 3 hours beginning with the first drop in receiving graduate. Calculate volume of oil absorbed by the sample.

4.4.2.1 Calculation.

$$\text{Milliliter of oil absorbed per gram of sample} = \frac{a}{b}$$

where:

a = Difference between volume of oil collected in the receiving graduate and the volume put into the glass tube.

b = Weight of the sample in grams.

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4.4.3 **Absorption of distilled water.** Repeat the same procedure specified in 4.4.2, using distilled water at 25° ± 2°C. in lieu of the lubricating oil, with the glass tube set at an angle of 30° to the horizontal. Seventy five ml. of water shall be carefully poured (within a 15 second period) into the open end of the glass tube. Allow a draining period of 30 minutes beginning with the first drop received in the graduate. Calculate volume of water absorbed by the sample.

4.4.3.1 Calculation.

$$\text{Milliliter of water absorbed per gram of sample} = \frac{a}{b}$$

where:

a - Difference between volume of water collected in the receiving graduate and the volume put into the glass tube.

b = Weight of the sample in grams.

4.4.4 Solubility in distilled water. Dry 5 ± 0.1 g. of sample in an oven 102.5° ± 2.5°C. for 24 hours. Cool and weigh 2.00 ± 0.1 g. of the dried sample into a 250 ml. beaker. Add 100 ml. of distilled water (25° ± 2°C) and stir for 3 minutes. Filter the solution by decantation through a tared Gooch crucible, washing to residue several times with distilled water. Dry the crucible and residue at 102° ± 2.5°C. for 24 hours; cool and weigh. Calculate solubility in distilled water.

4.4.4.1 Calculation.

$$\frac{\text{Wt. of sample - wt. of dried residue}}{\text{wt. of sample}} \times 100 = \text{percent solubility in distilled water}$$

4.4.5 Resistance to attrition.

4.4.5.1 Equipment and materials.

Sieve shaker - Ro-Tap or other apparatus which imparts a gyratory motion to a nest of sieves.

Sieves - U. S. Standard, Nos. 10, 25, 80, and bottom pan.

Steel balls - Approximately 1/4 inch diameter, requires about 300 balls having a total weight of 355 ± 5 g.

Balance - 0.05 g. sensitivity.

4.4.5.2 Procedure. Prepare a sample of the material to be tested which will pass through a 10-mesh screen but be retained on a 25-mesh screen by screening about 100 g. of the absorbent material by placing it on a sieve shaker for 5 minutes. Use 50.0 ± 0.1 g. of this fraction in a bottom pan, along with 300 1/4-inch steel balls. Shake the pan and contents on the sieve shaker for exactly 20 minutes. After shaking, pass the contents of the pan through a 10-mesh screen. The steel balls are retained on the screen and removed. The size distribution of the sample is then determined by the procedure given above (five minutes on a sieve shaker), using the 25 mesh screen and the 80-mesh screen and the 50-g. sample. The fraction of the sample passing the 80-mesh screen is weighed. The percent breakdown is then calculated as follows:

$$\text{Breakdown, percent thru 80-mesh screen} = \frac{\text{quantity thru 80-mesh screen} \times 100}{50}$$

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Fifty (50) pounds of absorbent material shall be packed in a sack conforming to UU-S-48, sack No. 8 or 8X and moisture barrier grade 1. The sack shall meet the general feature requirements listed for level A packing in Table XII. The sack shall be closed in accordance with the appendix of the sack specification.

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5.1.2 Level B. Fifty (50) pounds of absorbent material shall be packed in a sack conforming to UU-S-48, sack No. 4 or 4X. The sack shall meet the general feature requirements listed for level B packing in Table XII. The sack shall be closed in accordance with the appendix of the sack specification.

5.1.3 Level C. Absorbent material in quantities as specified (see 6.2) shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with the Uniform or National Motor Freight Classification.

5.2 Marking. In addition to any special markings required in the procurement documents, marking of the shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies, or MIL-STD-129 for Military agencies, as applicable (see 6.).

6. NOTES

6.1 Intended use. The absorbent material covered by this specification is intended for use on floors and decks of concrete, wood, and steel to absorb oil or mixtures of oil and water.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable level of packing required (see 5.1).

6.3 Basis of purchase. Absorbent material should be purchased by net weight.

Military Interest:

The Defense Department has waived coordination on all revisions or amendments of this specification until further notice.

Preparing Activity:

GSA-FSS

CIVIL AGENCIES INTEREST:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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